

COMPUTER PROGRAMMING CONCEPTS (390)

—OPEN EVENT—

STATE – 2019

DO NOT WRITE ON TEST BOOKLET

TOTAL POINTS _____ *(100 points)*

Failure to adhere to any of the following rules will result in disqualification:

- 1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.**
- 2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.**
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No more than sixty (60) minutes testing time

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MULTIPLE CHOICE

Identify the choice that best completes the statement or answers the question. Mark “A” if the statement is true. Mark “B” if the statement is false.

1. _____ refers to the process of hiding the internal details of objects and their methods.
 - a. Abstraction
 - b. Encapsulation
 - c. Assumption
 - d. Polymorphism
2. In a problem statement, you can limit complexity by making _____?
 - a. constants
 - b. variables
 - c. assumptions
 - d. objects
3. Which of the following is a scripting language?
 - a. Perl
 - b. PHP
 - c. Ruby
 - d. all of the above
4. A(n) _____ changes the order in which instructions are carried out by directing the computer to execute an instruction elsewhere in the program.
 - a. programming paradigm
 - b. sequence control structure
 - c. formal method
 - d. sequential execution
5. What third-generation languages were used extensively for business and scientific applications?
 - a. COBOL
 - b. Fortran
 - c. C
 - d. All the above
6. What is the term that refers to the sequence in which a computer executes program instructions?
 - a. flow control
 - b. walkthrough
 - c. control structure
 - d. step-through

7. What is a factor that remains the same throughout a program?
 - a. object
 - b. constant
 - c. variable
 - d. algorithm
8. What is a graphical representation of the way a computer should progress from one instruction to the next when it performs a task?
 - a. pseudocode
 - b. paradigm
 - c. VDE
 - d. flowchart
9. _____ is the ability (in programming) to present the same interface for differing underlying forms (data types).
 - a. inheritance
 - b. polymorphism
 - c. encapsulation
 - d. abstraction
10. Which of the following is *not* a characteristic for a good problem statement?
 - a. It clearly specifies the known information.
 - b. It specifies who is affected.
 - c. It contains detailed descriptions of the processes and tools that need to be developed.
 - d. It specifies any assumptions that define the scope of the problem.
11. Which of the following types of programmers specialize in developing system software such as device drivers, security modules, and communications software?
 - a. Application programmers
 - b. Database Application programmers
 - c. Web Development programmers
 - d. Network System programmers
12. What defines the variables in a class?
 - a. function
 - b. control structure
 - c. class attributes
 - d. class methods
13. C++, Visual Basic, C#, Python, and Java are all examples of what kind of language?
 - a. assembly
 - b. machine
 - c. mnemonic
 - d. programming
14. Pseudocode is standardized, each programmer writes pseudocode the same.
 - a. True
 - b. False

15. What is the output of the following algorithm if AMOUNT is equal to \$250?

```
if AMOUNT is greater than $500
    print "That's too much."
else if AMOUNT is greater than $200
    print "Still too much."
else if AMOUNT is greater than $100
    print "That's about right."
else
    print "Amount is too low."
End if
```

- a. That's too much.
 - b. That's about right.
 - c. Still too much.
 - d. Amount is too low.
16. In C++ and Java, the float and double variable data types are used to store real numbers.
- a. True
 - b. False
17. Which of the following data types can contain a real number?
- a. bool
 - b. char
 - c. float
 - d. int
18. Which of the following translates all of a program's high-level instructions into machine language before executing the program?
- a. interpreter
 - b. translator
 - c. executor
 - d. compiler
19. Which development component combines the object file with other machine code necessary for C++ or Java programs to run correctly?
- a. object code
 - b. linker
 - c. IDE
 - d. source code
20. Comparison operators evaluate to what type of value?
- a. Boolean
 - b. character
 - c. integer
 - d. real

21. What is the value of m after the following fragment of pseudocode is executed?

```
a = 2
b = 2
c = 4
if (a > b)
    if (a > c)
        m = 1
    else
        m = 2
    End if
else
    if (c != 0)
        m = 3
    else
        m = 4
    End if
End if
```

- a. 1
- b. 2
- c. 3
- d. 4

22. A conditional or decision node in a flowchart is represented by which shape?

- a. rectangle
- b. diamond
- c. oval
- d. parallelogram

23. An overflow error occurs when _____.

- a. a decimal value is stored into an integer-typed variable
- b. an integer is divided by a floating-point number
- c. a value is too large for its data type
- d. too many variables are defined in a single program

24. Which numerical base includes the use of letters to express numerical values?

- a. Binary
- b. Octal
- c. Decimal
- d. Hexadecimal

25. Dividing a problem or task into parts is an example of _____.

- a. Generality
- b. Abstraction
- c. Encapsulation
- d. Compilation

26. What is the value of *j* after following loop terminates?

```
int j = 0
for (int x = 0; x < 10; x += 2)
    j += 1
```

- a. 0
- b. 1
- c. 5
- d. 10

27. Which line of code takes the value count and adds three to its current value?

- a. count = 3++
- b. count = ++3
- c. count = +3
- d. count += 3

28. What is the process of locating and correcting the errors in a program called?

- a. defragging
- b. debugging
- c. deflating
- d. extrapolating

29. What is the term used for the assignment of a beginning value to a memory location?

- a. revision
- b. implicit type conversion
- c. explicit type conversion
- d. initializing

30. Examine the following code. What type of error do you find and what is the correct line of code if the goal of the program is to calculate a grade point average from the point values of three classes?

```
int gpa
int numclasses = 3
int class1 = 4
int class2 = 2
int class3 = 3
gpa = (class1 + class2) + class3/numclasses
```

- a. syntax error, gpa += (class1 + class2) + class3/numclasses
- b. syntax error, gpa = class1++class2++class3++/numclasses
- c. logic error, gpa = (class1 + class2 + class2) / numclasses
- d. logic error, gpa = class1 + class2 + (class3/ numclasses)

31. The following set of statements is an example of what type of structure?

```
if time-of-day is greater than noon
    set afternoon access permissions
else
    set regular access permissions
end if
```

- a. selection
- b. sequence
- c. iterative
- d. infinite

32. An accountant that is part of a programming team working on a program that requires accounting experience is an example of which a(n) _____.

- a. drag
- b. engineer
- c. subject matter expert
- d. top-level coordinator

33. After execution of the following code, what is the value of *num* if the input value is 6?

```
int num = getValue
if (num > 0)
    num = num + 10
else
    if (num >= 5)
        num = num + 15
```

- a. 5
- b. 6
- c. 16
- d. 21

34. What is the *output* of the following code?

```
int num = 10
while (num > 10)
    num = num - 2
output num
```

- a. 0
- b. 8
- c. 10
- d. None of these

35. The following code is an example of a(n) _____ loop.

```
Boolean found = true
while (found)
    entry = get input
    triple = entry * 3
    if (entry > 33)
        found = false
```

- a. flag-controlled
- b. counter-controlled
- c. EOF-controlled
- d. remote-controlled

36. In the code below, what message will be printed to the screen the most times, “Red” or “Blue”?

```
for (int x = 1; x <=3; x++)
    output “Red”
for (int y = 1; y <=3; y++)
    output “Blue”
```

- a. Red
- b. Blue
- c. Both will be printed an equal number of times
- d. Neither will be printed to the screen

37. Suppose x is 5 and y is 7. What is the value of the following pseudocode expression?

(x not equal 7) AND (x less than or equal to y)

- a. true
- b. false
- c. This is an invalid expression
- d. None of these

38. What is the value of the following expression: $9 * 2 - 8 > 5 + 2 / 2$?

- a. 10
- b. 6
- c. True
- d. False

39. What is the term for a unique number that identifies each item in a collection?

- a. accumulator
- b. counter
- c. tracker
- d. index

40. What is the *output* of the code below?

```
if ( 6 is greater than 8)
    output ("**")
    output ("****")
else if ( 9 equals 4)
    output ("****")
else
    output ("**")
```

- a. *
- b. **
- c. ***
- d. ****

41. In a *for* loop, which of the following is executed first?

- a. update expression
- b. logical expression
- c. initial expression
- d. for loop statement

42. Which of the following terms indicates assigning a beginning value to a counter or accumulator?

- a. Introducing
- b. Incrementing
- c. Updating
- d. Initializing

43. What is the term for a numeric variable that is used to add something together?

- a. counter
- b. updater
- c. incrementor
- d. accumulator

44. What of the following stores the information passed to the procedure when the procedure is invoked?

- a. Events
- b. Variables
- c. Constraints
- d. Parameters

45. _____ case is utilized when you capitalize the first letter in the procedure name and the first letter of each subsequent word in the name?

- a. Alto
- b. Camel
- c. Pascal
- d. Object

46. Which of the following standard flowchart symbols is used for calculations?
- Oval
 - Rectangle
 - Parallelogram
 - Diamond
47. Which of the following statements shows the *correct* order of logical operations?
- The *not* operator is first, the *and* operator is next, then the *or* operator.
 - The *and* operator is first, the *not* operator is next, then the *or* operator.
 - The *not* operator is first, then the *or* operator, then the *and* operator.
 - The *not* operator is first, the *and* and the *or* operators are of equal order.
48. What is the binary equivalent of the decimal number 7?
- 1001
 - 0111
 - 0011
 - 1010
49. Which of the following would be a valid name for a memory location?
- my.name
 - 3oreos
 - a1B2c3D4
 - me&you
50. Which of the following *for* clauses is correct for a generic *for* clause?
- for (int x = 1; x < 10; x += 1)
 - for (x < 10; x = 1; x += 1)
 - for (int x = 1; x += 1)
 - for (int x = 1)